



Modelling Financial Markets

Chapter 18: Systemic Risk and Networks



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CATFIN: CATastrophic risk in the FINancial sector
- ▣ CDS: Credit Default Swap
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ ECB: European Central Bank
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ FDIC: Federal Deposit Insurance Corporation
- ▣ FDR: False Discovery Rate
- ▣ FEVD: Forecast Error Variance Decomposition
- ▣ FRM: Financial Risk Meter
- ▣ GAAP: Generally Accepted Accounting Principles (US)
- ▣ GACV: Generalized Approximate Cross-Validation
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HSBC: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- ▣ IFRS: International Financial Reporting Standards
- ▣ ING: ING Groep (Internationale Nederlanden Groep) (the Dutch banking group ING)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator

Acronyms used in this material (2/2)

- ▣ LRMES: Long-Run Marginal Expected Shortfall
- ▣ MES: Marginal Expected Shortfall
- ▣ OLS: Ordinary Least Squares
- ▣ QFVAR: Quantile Factor Vector AutoRegression
- ▣ QML: Quasi-Maximum Likelihood
- ▣ RTD: Relative Tail Dependence
- ▣ SG: Société Générale (the French banking group Société Générale)
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ SVB: Silicon Valley Bank
- ▣ TENET: Tail-Event driven NETWORK risk
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VIX: CBOE Volatility Index

Systemic Risk and the Route of This Chapter

From one bank to the system

- **Systemic risk:** the risk that the distress of one or several institutions impairs the financial system as a whole, and through it the real economy
- An **externality:** a bank does not pay for the losses its failure imposes on others
- Three families of measures Billio, Getmansky, Lo & Pelizzon (2012); survey Benoit, Colliard, Hurlin & Pérignon (2017)
 - ▶ **Exposure:** what does the bank lose when the market crashes? MES, SRISK
 - ▶ **Contribution:** how much worse is the system when the bank is in distress? ΔCoVaR (CoVaR: the Value-at-Risk, VaR, of the system conditional on the bank)
 - ▶ **Connectedness:** through which links do shocks travel? Spillover and tail networks

Route of this chapter

- Part I: data and conventions; exposure (MES, LRMEs, SRISK); contribution (CoVaR, ΔCoVaR) and how far rankings can be trusted
- Part II: spillover networks, the Financial Risk Meter, stress tests and climate risk, a case study on quantile connectedness, one pass of the AI discovery loop
- Proofs, estimation details and robustness checks sit in the **Appendix** at the end
- Textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017)

Bank Runs: Northern Rock 2007 and Silicon Valley Bank 2023



Depositors queue outside a Northern Rock branch,
Brighton, 14 September 2007

Photo: [Dominic Alves \(2007\)](#); CC BY 2.0; [Wikimedia](#)

[Commons](#)

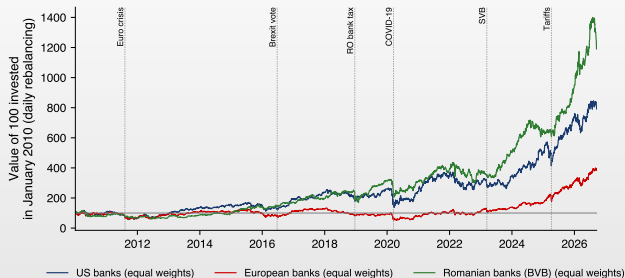
- Northern Rock (2007): a British mortgage lender hit by the freeze in funding markets after the US subprime losses; depositors queued to withdraw their savings
- SVB (2023): a digital run; the FDIC took the bank over on 10 March [FDIC \(2023\)](#); two days later all depositors of SVB and Signature Bank were protected on systemic risk grounds [Federal Reserve Board \(2023\)](#)
 - ▶ The stress-testing section measures the loss of a bank portfolio in that window
- Channels of contagion
 - ▶ **Funding runs:** depositors and lenders withdraw from any bank that looks similar
 - ▶ **Common exposures and fire sales:** forced selling lowers prices for every holder
 - ▶ **Counterparty links:** one bank's liability is another bank's asset, so a default passes losses to its creditors

Data and Conventions

Group	Series	Market index
US banks	JPMorgan, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo	S&P 500
European banks	Deutsche Bank, BNP Paribas, Santander, ING, HSBC	Euro Stoxx 50
Romanian banks	Banca Transilvania, BRD-Groupe Société Générale	BET
Volatility	the VIX index (implied volatility of the S&P 500)	

- Daily prices from January 2010 to 18 September 2026, adjusted for dividends and splits; 3 960 common trading days
- Prices are aligned on common days first, then daily log returns $r_t = 100(\ln P_t - \ln P_{t-1})$, in %
- Romanian banks: days without trades removed; historical stress scenarios for the international banks start in 2007
- Return X , loss $L = -X$; the level α is a **tail probability**: VaR 1%, CoVaR 1%, MES 5% ([Chapter 7](#))
 - ▶ $\text{VaR}_\alpha(X) = -q_\alpha(X)$, with q_α the α -quantile; VaR, CoVaR and MES are positive numbers when they are losses
- **System return** X^{sys} : the equal-weighted portfolio of the other banks of the same region
 - ▶ $X^{\text{sys}} = \frac{1}{n} \sum_j X_j$ averages daily log returns: a linear aggregate, a first-order approximation of the log return of the daily rebalanced portfolio; MES, CoVaR and the Euler identity are stated for this aggregate

Three Banking Systems, 2010–2026

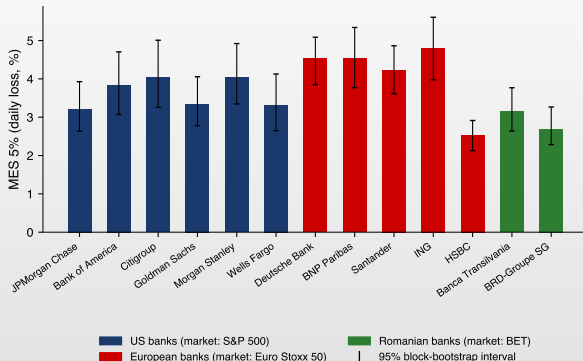


- ▣ Equal-weighted portfolios of the banks of each region, rebalanced daily; vertical lines mark the episodes used in this chapter
- ▣ 100 invested in January 2010 became 791 in US banks, 384 in European banks and 1233 in Romanian banks
 - ▶ Largest peak-to-trough fall: US 51% (2011-11-23), Europe 61% (2020-04-21), Romania 47% (2012-06-06)
- ▣ Annualised volatility of the portfolios: 29%, 30% and 23%, against 18% (S&P 500), 21% (Euro Stoxx 50) and 18% (BET)
- ▣ The falls are **simultaneous**: the euro crisis, COVID-19 and the 2023 bank failures hit all regions in the same weeks
- ▣ Romanian banks also have their own shock: the bank-tax announcement of December 2018

Marginal Expected Shortfall and the Euler Property

- **MES** (Marginal Expected Shortfall) of bank i at level α Acharya, Pedersen, Philippon & Richardson (2017)
 - ▶ $MES_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$
 - ▶ The average loss of the bank on the worst α share of days for the market m
- Why “marginal”: if the conditioning variable is the portfolio itself, $m = \sum_i w_i X_i$, then $ES_\alpha(X_m) = \sum_i w_i MES_{i,\alpha}$ (Euler allocation; proof in the [Appendix](#))
 - ▶ w_i : portfolio weights; $ES_\alpha(X_m) = -\mathbb{E}[X_m \mid X_m \leq q_\alpha(X_m)]$: the average loss of m on its own worst α days
 - ▶ Our markets (S&P 500, Euro Stoxx 50, BET) are **not** portfolios of these 13 banks: here MES measures **exposure**, not a share of the index ES
- Check on our data (six US banks, $\alpha = 5\%$, 198 tail days): portfolio $ES_{5\%} = 4.32\% =$ average MES conditional on the portfolio (4.32%); conditioning on the S&P 500 instead gives 3.63%
- Historical estimate: average $-X_i$ over the days with $X_m \leq \hat{q}_\alpha(X_m)$; with $\alpha = 5\%$ and 3960 days, about 200 days
- Market m : S&P 500 for US banks, Euro Stoxx 50 for European banks, BET for Romanian banks

MES 5% of Thirteen Banks



- Average daily loss of each bank on the worst 5% of days for its market index, 2010–2026; 95% intervals from a moving-block bootstrap (blocks of 20 days)
- US banks: from 3.22% (JPMorgan) to 4.04% (Morgan Stanley); European banks up to 4.81% (ING)
 - ▶ On the same days the market lost on average 2.72% (S&P 500) and 3.16% (Euro Stoxx 50): banks lose more than the market
- HSBC is the exception in Europe (2.54%, below the index): a London listing with Asian business
- Banca Transilvania 3.14% and BRD 2.69% on BET crash days; BET itself lost 2.70%
- Overlapping marginal intervals are not a test of equality: bank-to-bank differences need a paired test on joint bootstrap draws (Section 3)

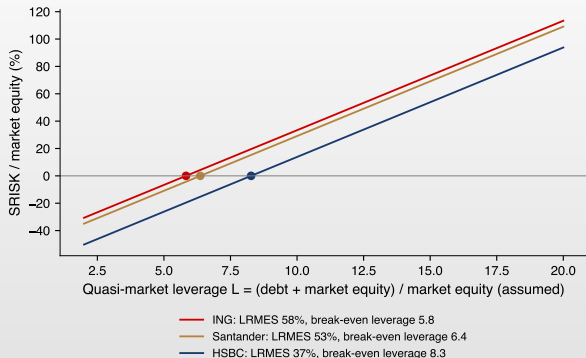
From MES to a Crisis: LRMES

- **LRMES** (Long-Run MES): the expected fall of the bank's equity in a crisis, a market fall below C over h days
 - ▶ $\text{LRMES}_{i,t} = -\mathbb{E}_t [R_{i,t+1:t+h} \mid R_{m,t+1:t+h} < C]$, arithmetic h -day returns
 - ▶ $R_{i,t+1:t+h}$, $R_{m,t+1:t+h}$: returns of bank i and of the market from $t+1$ to $t+h$; C : the crisis threshold; $\mathbb{E}_t$: expectation given the information at t
 - ▶ [Acharya, Engle & Richardson \(2012\)](#) (Section I): $C = -40\%$ over six months; [Brownlees & Engle \(2017\)](#) (Section 1.1): $C = -10\%$ over $h = 22$ days
- Shortcut [Acharya, Engle & Richardson \(2012\)](#), Section I, for “versions of the model where the simulation is not yet implemented”: $\text{LRMES}_i \approx 1 - \exp(-18 \times \text{MES}_i^{2\%})$
 - ▶ $\text{MES}_i^{2\%}$: average daily loss of the bank on days when the market falls by more than 2% (in decimals); example $\text{MES}^{2\%} = 3\%$ gives $1 - e^{-0.54} = 41.7\%$
 - ▶ The 18 is a calibration constant; Seminar 18, A4 derives where it comes from
- Shortcut on our banks: from 37% (HSBC) to 58% (ING); JPMorgan 48%, Banca Transilvania 52%
 - ▶ The market fell by more than 2% on 138 days in the US, 196 in the euro area and 111 on the BVB

SRISK and the Break-Even Leverage

- A bank must keep equity of at least a share k of its quasi-assets (book debt D plus market equity W) [Brownlees & Engle \(2017\)](#), Section 1.1, Eq. (1)
 - ▶ $SRISK_i = k D_i - (1 - k)(1 - LRMES_i) W_i$, assuming debt is not renegotiated in the crisis
 - ▶ $k = 8\%$ for US firms [Brownlees & Engle \(2017\)](#); $k = 5.5\%$ for European firms, whose IFRS balance sheets report larger assets than US GAAP [Engle, Jondeau & Rockinger \(2015\)](#)
- Positive SRISK = shortfall; the SRISK of the system is the sum of the positive values; [V-Lab, New York University Stern School of Business](#) publishes it for large banks with their balance sheets
- Divide by W and write the quasi-market leverage $L = (D + W)/W$
 - ▶ $\frac{SRISK}{W} = k(L - 1) - (1 - k)(1 - LRMES)$
- SRISK is positive once leverage exceeds the break-even leverage
 - ▶ $L^* = 1 + \frac{(1 - k)(1 - LRMES)}{k}$
- Example: $D = 900$, $W = 100$, $LRMES = 41.7\%$
 - ▶ $SRISK = 0.08 \times 900 - 0.92 \times (1 - 41.7\%) \times 100 = 72 - 53.6 = 18.4$
- Example above: $L^* = 7.7$, while the bank has $L = 10$: a shortfall
- Our data contain prices, not balance sheets: we report L^* from the estimated LRMES and treat leverage as an assumption

SRISK as a Function of Leverage



- Banks with the highest, median and lowest LRMES; leverage on the horizontal axis is assumed, not observed; dots mark L^*
- With $k = 8\%$: $L^* = 5.8$ for ING, 6.4 for Santander, 8.3 for HSBC
 - ▶ US: JPMorgan 7.0, Morgan Stanley 6.0; Romania: Banca Transilvania 6.5, BRD 7.1
- A bank whose debt is more than about 5–7 times its market equity would need new capital in a 40% market crash
- With $k = 5.5\%$ the thresholds rise (ING 8.2, HSBC 11.9): the prudential ratio is a policy choice, and SRISK depends on it
- Limits: LRMES is an extrapolation from daily data, and market equity itself falls in a crisis

LRMES: GJR–GARCH, DCC and Simulation (1/2): the Model

- Model of [Brownlees & Engle \(2017\)](#), Section 1.2: zero-mean log returns of bank and market, GJR–GARCH(1,1) volatilities (GARCH: generalized autoregressive conditional heteroskedasticity) [Glosten, Jagannathan & Runkle \(1993\)](#) and a DCC(1,1) correlation [Engle \(2002\)](#), estimated by two-step quasi-maximum likelihood (QML)
- Volatility, for the bank and for the market: today's variance reacts to yesterday's squared return, more after a fall
 - ▶ $\sigma_t^2 = \omega + (\alpha + \gamma 1\{r_{t-1} < 0\}) r_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ σ_t^2 : conditional variance of the return r_t ; $\omega > 0$: constant; α : reaction to a shock; $\gamma > 0$: extra reaction after a fall (leverage effect); β : persistence; $1\{\cdot\}$: 1 if the condition holds, else 0
- Correlation: a GARCH-type recursion for the matrix Q_t , rescaled to a correlation
 - ▶ $Q_t = (1 - a - b) S + a \epsilon_{t-1} \epsilon_{t-1}^\top + b Q_{t-1}$, $\rho_t = Q_{12,t} / \sqrt{Q_{11,t} Q_{22,t}}$
 - ▶ $\epsilon_t = (r_{it}/\sigma_{it}, r_{mt}/\sigma_{mt})^\top$: standardised returns; S : their sample correlation matrix; a : reaction, b : persistence of the correlation; ρ_t : today's bank–market correlation

LRMES: GJR-GARCH, DCC and Simulation (2/2): Simulation

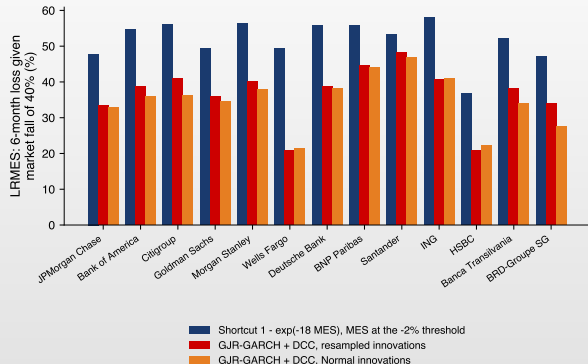
- Simulation (Brownlees & Engle, ▶ Appendix A; details in the ▶ Appendix of this deck): resample **pairs** of standardised innovations $(\xi_{it}, \epsilon_{mt})$, filter h days ahead from today's σ^2 and ρ , keep the paths with $R_{m,t+1:t+h} < C$
 - ▶ LRMES = minus the average bank return over the crisis paths; resampling pairs keeps the tail dependence that the Normal distribution misses
- Simulation on data to 2026-09-18; 200 000 paths of 22 days per market ($C = -10\%$), 400 000 paths of 126 days ($C = -40\%$)
 - ▶ Probability of the 22-day crisis today: 1.3% (S&P 500), 4.3% (Euro Stoxx 50), 4.0% (BET); on 31 March 2020: 12%, 19%, 8%

LRMES by Simulation: Thirteen Banks

Bank	LRMES today	90% range	LRMES 31.03.2020	ρ_T	L^*
JPMorgan	10.6	[-2, 25]	23.7	0.53	11.3
Bank of America	14.3	[-4, 35]	23.2	0.44	10.9
Citigroup	13.0	[-1, 28]	32.8	0.54	11.0
Goldman Sachs	16.5	[1, 34]	25.0	0.65	10.6
Morgan Stanley	16.3	[1, 32]	27.2	0.65	10.6
Wells Fargo	6.7	[-8, 24]	22.9	0.32	11.7
Deutsche Bank	14.4	[2, 28]	25.7	0.66	10.8
BNP Paribas	15.6	[2, 33]	28.5	0.68	10.7
Santander	16.6	[3, 34]	26.4	0.76	10.6
ING	12.5	[1, 27]	32.9	0.68	11.1
HSBC	8.3	[-2, 19]	10.9	0.58	11.5
Banca Transilvania	12.4	[-2, 34]	17.2	0.74	11.1
BRD-Groupe SG	11.4	[-2, 28]	15.8	0.64	11.2

- LRMES (%) for a market fall below -10% in 22 days [Brownlees & Engle \(2017\)](#); 90% range: 5% and 95% quantiles of the bank's loss over the crisis paths; ρ_T : last DCC correlation; L^* with $k = 8\%$
- LRMES is conditional: it moves with today's volatility and correlation
 - JPMorgan: 10.6% today, 23.7% with the information of 31 March 2020; Deutsche Bank 14.4% and 25.7%
- The 90% range is wide: LRMES is an average over crisis paths, and the capital shortfall of a single crisis can be much larger [Brownlees & Engle \(2017\)](#), Eq. (2)
- With a one-month, -10% crisis, LRMES is about a third of the six-month, -40% numbers of [Acharya, Engle & Richardson \(2012\)](#): the scenario, not only the bank, sets the level

Model Risk: LRMES under Three Specifications



- Six-month crisis, market fall of 40% [Acharya, Engle & Richardson \(2012\)](#); shortcut $1 - e^{-18\text{MES}^{2\%}}$ vs simulation from GJR-GARCH + DCC with resampled or Normal innovations, data to 2026-09-18
- Simulated LRMES divided by the shortcut: between 0.42 and 0.90 across the 13 banks; rank correlation 0.77
 - ▶ JPMorgan: 48% (shortcut), 33% (resampled), 33% (Normal innovations); ING: 58%, 41%, 41%
- The shortcut uses the whole 2010–2026 history; the simulation starts from today's volatility and correlation
- Resampled vs Normal innovations: rank correlation 0.95; the innovation distribution changes levels more than rankings
- Same banks, same data, different defensible models, different numbers: the “model risk of risk models” [Danielsson, James, Valenzuela & Zer \(2016\)](#)

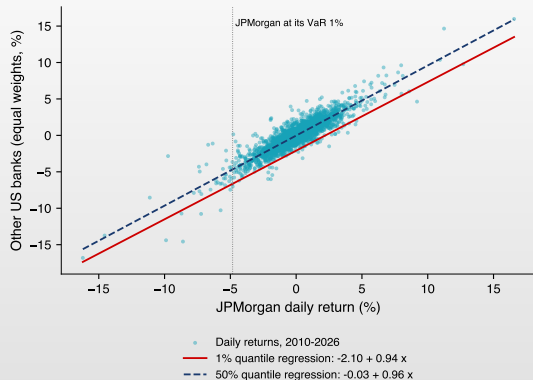
CoVaR: the VaR of the System Given a Bank in Distress

- **CoVaR** of the system conditional on bank i at its VaR [Adrian & Brunnermeier \(2016\)](#); X^{sys} , X_i : returns of the system and of bank i
 - ▶ $P\left(X^{sys} \leq -\text{CoVaR}_\alpha^{sys|i} \mid X_i = -\text{VaR}_{i,\alpha}\right) = \alpha$
 - ▶ The loss of the system that is exceeded with probability α on a day when bank i is at its own VaR
- **ΔCoVaR** : the change from the median state of the bank to its distress state
 - ▶ $\Delta\text{CoVaR}_\alpha^{sys|i} = \text{CoVaR}_\alpha^{sys|X_i=q_\alpha} - \text{CoVaR}_\alpha^{sys|X_i=q_{50\%}}$
 - ▶ $\text{CoVaR}_\alpha^{sys|X_i=q}$: CoVaR with the bank return fixed at its quantile q ; $q_{50\%}$: the median, a normal day
- ΔCoVaR measures the **contribution** of bank i to the tail risk of the system
- Reversing the roles gives the **exposure CoVaR**: the VaR of bank i when the system is in distress

Estimating CoVaR by Quantile Regression

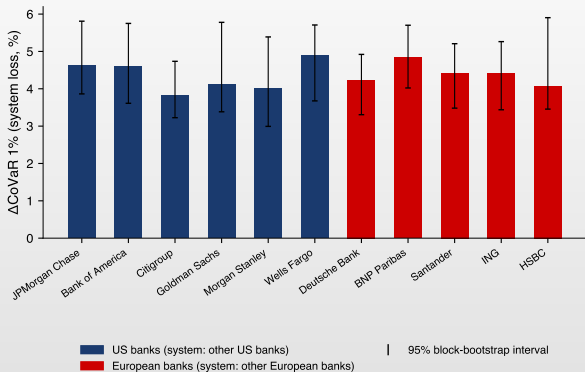
- The α -quantile of Y given x is linear: $q_\alpha(Y | x) = a_\alpha + b_\alpha x$ [Koenker & Bassett \(1978\)](#)
 - ▶ Here Y : the system return, x : the bank return; on the data y_t, x_t : $(\hat{a}_\alpha, \hat{b}_\alpha) = \arg \min_{a,b} \sum_t \rho_\alpha(y_t - a - bx_t)$,
 $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
 - ▶ ρ_α is the **check function**: it weights negative residuals by $1 - \alpha$ and positive ones by α ; $\alpha = 50\%$ gives the median regression
- 1. Quantile regression of the system on the bank at level α
 - ▶ $\hat{q}_\alpha(X^{sys} | X_i) = \hat{a}_\alpha + \hat{b}_\alpha X_i$
- 2. Plug in the bank's own quantiles: $\hat{q}_\alpha(X_i)$ (distress) and $\hat{q}_{50\%}(X_i)$ (median)
 - ▶ $\text{CoVaR}_\alpha^{sys|i} = - \left(\hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i) \right)$
- 3. Take the difference
 - ▶ $\Delta \text{CoVaR}_\alpha^{sys|i} = \hat{b}_\alpha (\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$
 - ▶ Only the slope matters: the intercept cancels
- Uncertainty: moving-block bootstrap of the days (blocks of 20 days keep volatility clusters together), 999 resamples

Quantile Regression: JPMorgan and the Other US Banks



- Daily returns 2010–2026 with the 1% and 50% quantile regression lines; the dotted line marks JPMorgan at its VaR 1%
- 1% line: $\hat{q}_{1\%} = -2.10 + 0.94 X_i$; median line: $-0.03 + 0.96 X_i$
 - Similar slopes: the 1% line is shifted down by the tail of the other banks, not steeper
- JPMorgan's VaR 1% is 4.83%; on such a day the other US banks have CoVaR 1% = 6.64%
 - Unconditional VaR 1% of the system: 5.07%; with JPMorgan at its median: 2.03%
- $\Delta \text{CoVaR}_{1\%} = 4.62\%$: JPMorgan in distress adds about 4.62 percentage points to the 1% loss of the other US banks
- This is association, not causation: a common shock hits JPMorgan and the others on the same day

ΔCoVaR 1% of Eleven Banks



- System: equal-weighted portfolio of the other banks of the same region; 95% moving-block bootstrap intervals
- All eleven banks lie between 3.81% (Citigroup) and 4.89% (Wells Fargo); the 95% intervals are 1.5–2.4 points wide
 - ▶ E.g. Citigroup [3.23, 4.74], Wells Fargo [3.68, 5.71]: the marginal intervals overlap
- Overlap is not a test: two estimates computed on the same days are dependent (in our bootstrap draws strongly positively correlated), so the interval for the difference can be narrower than either marginal interval
- At $\alpha = 5\%$ the values are smaller (JPMorgan 2.62%) and the ranking changes: Spearman correlation between the 1% and 5% rankings 0.10
 - ▶ A ranking that changes with α is not a robust basis for regulation

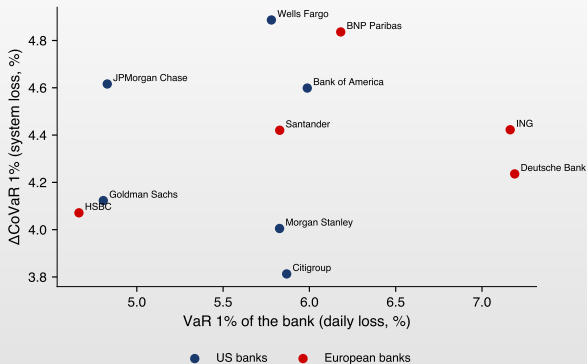
Testing ΔCoVaR 1%

- All tests use the same 999 moving-block bootstrap draws of days for every bank, so estimates of different banks keep their dependence
- (1) No contribution, $H_0: b_\alpha = 0$: Wald test with the sandwich variance of quantile regression [Koenker & Bassett \(1978\)](#)
 - ▶ Kernel (sandwich) t -ratios of $\hat{b}_{1\%}$ from 4.0 (HSBC) to 6.9 (JPMorgan): every bank contributes
- (2) The tail link equals the central link, $H_0: b_\alpha = b_{50\%}$: is the bank special in crises?
 - ▶ Rejected for 2 of 11 banks at 5% (2 after Holm): for most banks the tail link equals the central link
- (3) Equal contributions, $H_0: \Delta\text{CoVaR}_i = \Delta\text{CoVaR}_j$: the same resampled days for **all** banks give the joint distribution
 - ▶ Equal contributions within a region: 4 of 25 pairs differ at 5%, 0 after Holm; joint Wald $p = 0.01$ (US), 0.31 (Europe)
 - ▶ At 5%: 3 of 25 pairs differ, 0 after Holm; joint Wald $p = 0.014$ (US), 0.046 (Europe)
- Full table of slopes and p -values: [Appendix](#); significance and dominance tests for systemic importance: [Castro & Ferrari \(2014\)](#)

Losses at Least Equal to VaR: the Girardi–Ergün CoVaR

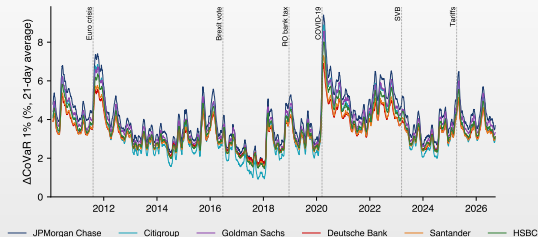
- Girardi & Ergün (2013) condition on the bank's loss being **at least** its VaR, not exactly equal to it
 - ▶ $P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq} \mid X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha \iff P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq}, X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha^2$
 - ▶ $\text{CoVaR}_{\alpha}^{\leq}$: CoVaR given a loss at least as large as the VaR; the joint form follows from $P(A \mid B) = P(A, B)/P(B)$ with $P(B) = \alpha$
 - ▶ Benchmark state: a one-standard-deviation event $|X_i - \mu_i| \leq \sigma_i$ (μ_i, σ_i : mean and standard deviation of X_i) instead of the median
- Why it matters
 - ▶ It includes more severe distress than the VaR day itself, and it is monotone in the dependence parameter
 - ▶ It can be **backtested**: the joint hit $\{X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq}, X_i \leq -\text{VaR}_i\}$ has probability α^2
- Historical estimate at $\alpha = 5\%$: the 5% quantile of the system on the 198 days with $X_i \leq \hat{q}_{5\%}(X_i)$
 - ▶ JPMorgan: $\text{CoVaR}_{5\%}^{\leq} = 7.70\%$ vs the quantile-regression $\text{CoVaR}_{5\%} = 3.81\%$; eleven banks: 7.3–8.2% vs 3.6–4.8%
 - ▶ About twice as large: averaging over the whole tail of the bank is a harsher conditioning event (Seminar 18, A2 derives this under the Normal distribution)

VaR and ΔCoVaR Measure Different Things



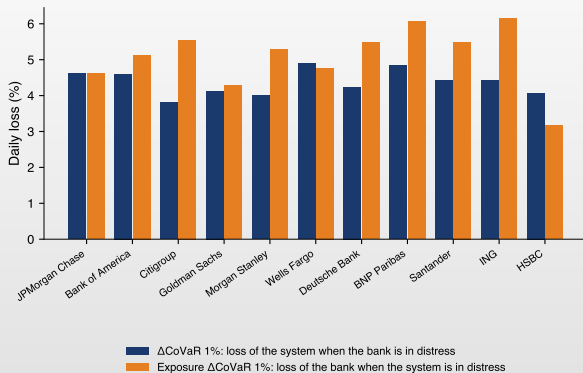
- Horizontal: VaR 1% of the bank; vertical: its ΔCoVaR 1%, eleven banks, 2010–2026
- Spearman rank correlation between VaR and ΔCoVaR : 0.16
 - ▶ Deutsche Bank and ING have the largest VaR (7.19%, 7.16%) but only average ΔCoVaR
 - ▶ JPMorgan has one of the smallest VaR values (4.83%) and one of the largest ΔCoVaR values
- This is the central message of [Adrian & Brunnermeier \(2016\)](#): regulating each bank by its own VaR ignores its contribution to the system
- Caveat: with eleven banks and overlapping intervals, the correlation itself is very uncertain (Seminar 18, B2)

ΔCoVaR 1% over Time



- ▣ Six banks, 21-day moving averages of the daily ΔCoVaR 1% estimated with lagged state variables
- ▣ State variables M_{t-1} : log VIX, the return of the regional index and of the system, all lagged one day [Adrian & Brunnermeier \(2016\)](#)
- ▣ Then $\Delta\text{CoVaR}_{\alpha,t} = \hat{b} \left(\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i) \right)$
- ▣ JPMorgan: 2.16% on average in 2017, 9.06% in March 2020; peak 10.99% on 2020-03-18
 - ▶ Deutsche Bank: 2.04% in 2017 and 6.65% in March 2020
- ▣ Market-based ΔCoVaR is low in calm periods, exactly when risk may be building up in balance sheets, which market data do not see
- ▣ [Adrian & Brunnermeier \(2016\)](#) therefore propose a forward ΔCoVaR , predicted from leverage, size and maturity mismatch

Contribution versus Exposure



- Blue: ΔCoVaR 1% (loss of the system when the bank is in distress); orange: exposure ΔCoVaR 1% (loss of the bank when the system is in distress)
- The direction matters: the two numbers differ for most banks
 - ▶ ING: contribution 4.42%, exposure 6.14%; BNP Paribas: 4.84% and 6.07%
 - ▶ HSBC: contribution 4.07%, exposure only 3.17%
- Exposure is larger than contribution for most European banks: the rise in the bank's tail loss when the system is in distress exceeds the rise in the system's tail loss when the bank is in distress; both are associations, not causal effects
- Exposure matters for the bank's own capital; contribution matters for the regulator of the system

How Much Can We Trust a Ranking?

- Rankings of banks by different systemic measures disagree, and each ranking has wide sampling uncertainty [Nucera, Schwaab, Koopman & Lucas \(2016\)](#)
 - ▶ They propose combining measures and reporting rank uncertainty instead of a single ordering
- Model risk: the same bank's risk measure varies widely across estimation methods [Danielsson, James, Valenzuela & Zer \(2016\)](#)
 - ▶ In this chapter: LRMES shortcut vs simulation (ratio 0.42–0.90), quantile-regression vs Girardi–Ergün CoVaR (about $\times 2$), $\alpha = 1\%$ vs 5% (Spearman 0.10)
- The effective tail sample is αn : 40 days at 1% and 198 at 5% out of 3 960
 - ▶ In our data the block-bootstrap standard error of $\Delta\text{CoVaR } 1\%$ is about 0.6 times the kernel-based standard error (median over banks): the two methods give clearly different uncertainty
- Out-of-sample backtests of $\text{CoVaR}_{5\%}^{\leq}$ reject for no bank, but with about 7.4 expected joint hits their power is low
(▶ [Appendix](#))
- What survives: large, tested differences (regions; the tail slope of some banks), not the order of eleven close point estimates

From Pairs to Networks: the Variance Decomposition (1/2)

- CoVaR looks at one bank and one system; a **network** looks at all pairs at once
 - ▶ Nodes: banks; directed edges: how much of the future uncertainty of bank i comes from shocks to bank j
- Vector autoregression (VAR) of order p for N series y_t : $y_t = c + \sum_{l=1}^p A_l y_{t-l} + \varepsilon_t$, $\text{Cov}(\varepsilon_t) = \Sigma$
 - ▶ Each series is regressed on p lags of all series: c : constants; A_l : $N \times N$ coefficient matrices; ε_t : shocks; Σ : their covariance matrix
- Variance-decomposition networks Diebold & Yilmaz (2009), Diebold & Yilmaz (2012), Diebold & Yilmaz (2014): the approach of this section
- Other networks: Granger causality Billio, Getmansky, Lo & Pelizzon (2012) (▶ [Appendix](#)), tail networks TENET Härdle, Wang & Yu (2016) and the Financial Risk Meter (next section)

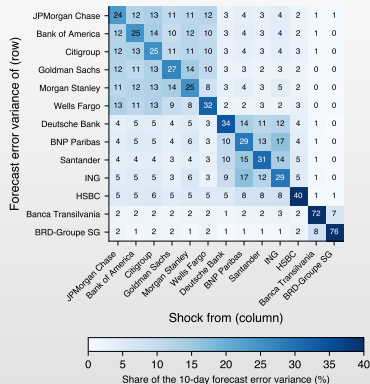
From Pairs to Networks: the Variance Decomposition (2/2)

- The H -day forecast error of y_i is a sum of shocks to all series: its variance can be split by source
 - ▶ Moving-average form $y_t = \mu + \sum_{h \geq 0} \Phi_h \varepsilon_{t-h}$: Φ_h is the response of y after h days to a shock today, $\Phi_0 = I$
- Orthogonal (Cholesky) shocks depend on the ordering of the series; the **generalized** decomposition does not [Koop, Pesaran & Potter \(1996\)](#), [Pesaran & Shin \(1998\)](#)
 - ▶ $\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$, normalised: $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$
 - ▶ Numerator: the part of the forecast error variance of i due to a shock in j ; denominator: the total forecast error variance of i
 - ▶ e_i : the i -th unit vector (it picks series i); σ_{jj} : the variance of shock j , the j -th diagonal element of Σ

The Connectedness Table and Range Volatility

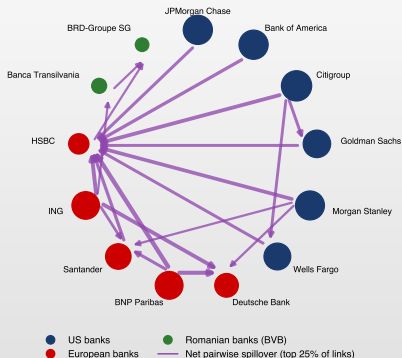
- $\tilde{\theta}_{ij}$ (in %): the share of the H -day forecast error variance of i due to shocks in j ; each row sums to 100 [Diebold & Yilmaz \(2014\)](#)
 - ▶ **From others:** $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$; **to others:** $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$
 - ▶ **Net:** $T_i - F_i$ (positive = net transmitter); **pairwise net:** $\tilde{\theta}_{ji} - \tilde{\theta}_{ij}$
- **Total connectedness:** $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$, the average share of variance that comes from others
 - ▶ Rows of θ_{ij} do not sum to one before normalisation because generalized shocks are correlated (Seminar 18, A5 derives this)
 - ▶ A pairwise net link $\tilde{\theta}_{ji} - \tilde{\theta}_{ij} > 0$ means that i transmits net to j and draws an arrow $i \rightarrow j$: the network of the next slides
- **Parkinson volatility** from the daily high H_t and low L_t [Parkinson \(1980\)](#)
 - ▶ $\hat{\sigma}_t^2 = \frac{(\ln H_t - \ln L_t)^2}{4 \ln 2}$, annualised $100\sqrt{252} \hat{\sigma}_t^2$, in logs
 - ▶ Unbiased for the variance of a Brownian motion without drift; more efficient than the squared return
- Our model: VAR on the log volatility of 13 banks, order chosen by the Bayesian information criterion ($p = 2$), horizon $H = 10$ days
 - ▶ Time zones: US banks close after European ones; the VAR lags capture the next-day transmission

The Connectedness Table of Thirteen Banks



- Row i , column j : share (%) of the 10-day forecast error variance of bank i due to shocks in bank j , full sample
- Total connectedness $C = 63.9\%$: on average two thirds of a bank's volatility risk comes from the other banks
 - On returns instead of volatility: $C = 75.4\%$
- Two regional blocks: a US bank receives 84.0% from US shocks and a European bank 72.7% from European shocks, own shock included
 - Across the Atlantic: a European bank receives 26.2% from US banks, a US bank 15.0% from European ones
- Romanian banks' volatility forecast errors come mostly from their own shocks: own share 72.3% (Banca Transilvania) and 76.1% (BRD); they receive 10.3% from US and 8.0% from European banks

The Net Spillover Network



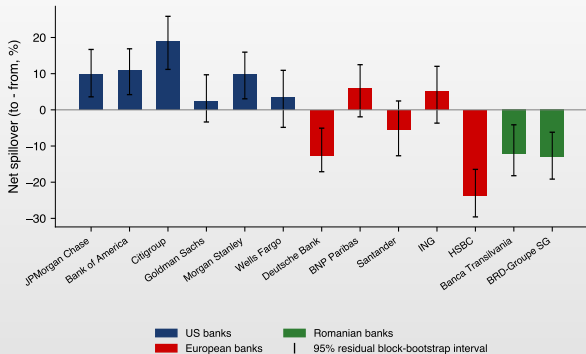
- Arrow from j to i when j transmits net to i (largest 25% of pairwise net spillovers); node size grows with the spillover to others
- Net transmitters: Citigroup (+19.0), Bank of America (+10.9), Morgan Stanley (+9.9), JPMorgan (+9.8)
 - ▶ To others: Citigroup 93.8%, JPMorgan 86.2% (the sum of a column can exceed 100)
- Net receivers: HSBC (−23.6), BRD (−13.0), Deutsche Bank (−12.6), Banca Transilvania (−12.2)
 - ▶ HSBC receives from almost every bank: it trades in London, between Asia and New York
- Within Europe, BNP Paribas and ING transmit to Deutsche Bank and Santander
- A full-sample network averages quiet and crisis years: the next chart lets it move

Connectedness over Time



- Top: total connectedness on moving 250-day windows (VAR(2), $H = 10$, every 5 days); bottom: net spillover summed by region
- The total index ranges from 47.4% (2024-07-24) to 79.3% (2020-08-27); mean 62.0%
 - ▶ COVID-19: from 61.8% in 2019 to 78.4% in spring 2020; it jumps within days and decays over a year
 - ▶ After the 2023 bank failures 66.0%; the calm year 2024 53.5%
- US banks are net transmitters in 89% of the windows (average net +48.4)
 - ▶ European banks were net transmitters during the euro crisis of 2011–2012 and mostly net receivers afterwards
- Romanian banks are net receivers on average (−24.1)

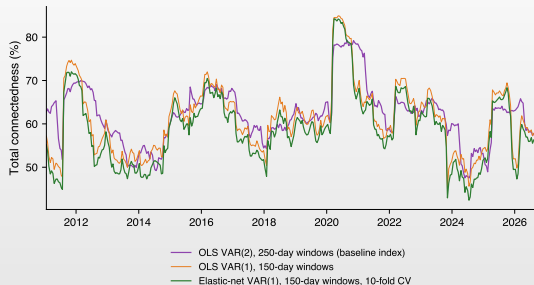
Estimation Error: Bootstrap Intervals for the Network



- A VAR(1) on 13 series has 169 slope coefficients; a 250-day window gives 250 observations per equation for 13 slopes, and the table is a nonlinear function of all of them
- Residual block bootstrap: resample blocks of 20 days of VAR residual vectors, rebuild the series with the estimated VAR, re-estimate, recompute the table; 999 draws
- Full sample, $p = 2$: $C = 63.9\%$ with 95% interval [59.9, 65.8]
 - ▶ Window of the peak (250 days to 2020-08-27, $p = 2$): $C = 79.3\%$, interval [65.9, 83.0]; bootstrap mean 75.8%, so small samples also bias C
- 8 of 13 net positions have intervals that exclude zero; for the other 5 the label “transmitter” or “receiver” is not identified by the data
- Report the interval next to every net number, and draw only arrows whose pairwise net link is significant

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OLS and Elastic-Net Connectedness in Rolling Windows



- [Demirer et al. \(2018\)](#), Eq. (9): each VAR equation by the adaptive elastic net, λ chosen equation by equation by 10-fold cross-validation
- Full sample, adaptive elastic net: $C = 63.3\%$ (OLS 63.9%); 59% of the coefficients are non-zero
- Means over 372 windows: 62.0% (baseline index: OLS VAR(2), 250-day windows), 61.1% (OLS, 150 days), 58.7% (elastic net, 150 days)
- The dynamics agree: correlation 0.82 between the elastic-net index and the baseline index; the crisis peaks are in the same weeks
- Levels depend on window, estimator and lag order; changes over time are the robust part of the index

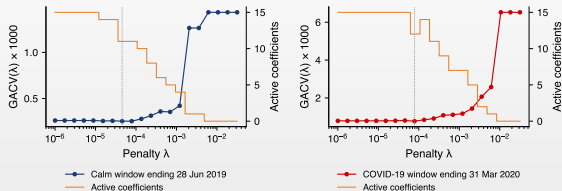
The Financial Risk Meter and the GACV Penalty (1/2)

- For each firm j , a quantile regression at $\tau = 5\%$ of its return on the returns of **all** other firms and on lagged macro variables Mihoci, Althof, Chen & Härdle (2020)
 - ▶ With 100 firms and a 63-day window there are more regressors than days: a penalty is needed
- **LASSO** (Least Absolute Shrinkage and Selection Operator) quantile regression Belloni & Chernozhukov (2011)
 - ▶ $\hat{\beta}_j(\lambda) = \arg \min_{\beta} \frac{1}{n} \sum_t \rho_{\tau} \left(X_{j,t} - \beta_0 - \beta^{\top} Z_{-j,t} \right) + \lambda \|\beta\|_1$
 - ▶ $X_{j,t}$: return of firm j ; $Z_{-j,t}$: returns of the other firms and the lagged macro variables; β_0 : intercept; ρ_{τ} : the check function; $\|\beta\|_1 = \sum_k |\beta_k|$; n : days in the window
 - ▶ λ switches off weak links: a larger λ leaves fewer active coefficients
- **GACV** (Generalized Approximate Cross-Validation) trades fit against the number of active coefficients
 - ▶
$$\text{GACV}(\lambda) = \frac{\sum_t \rho_{\tau} \left(X_{j,t} - \hat{X}_{j,t}(\lambda) \right)}{n - \text{df}(\lambda)}, \quad \hat{X}_{j,t}(\lambda): \text{fitted quantile; df} = \text{number of nonzero coefficients; the selected } \lambda_j \text{ minimises GACV}$$

The Financial Risk Meter and the GACV Penalty (2/2)

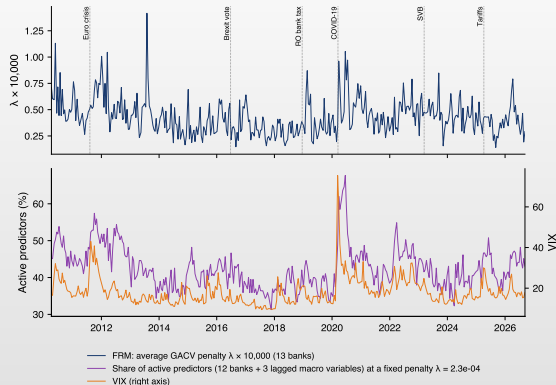
- The **FRM** (Financial Risk Meter) is the average of the selected λ_j over firms, designed as an indicator of tail interdependence and systemic stress
 - ▶ Built for US financial firms [Mihoci, Althof, Chen & Härdle \(2020\)](#), emerging markets [Ben Amor, Althof & Härdle \(2022\)](#) and China [Wang, Althof & Härdle \(2023\)](#)
- Our version: 13 banks, macro variables S&P 500, Euro Stoxx 50 and VIX at $t - 1$, 63-day windows every 10 days
 - ▶ 15 regressors for 63 days: far fewer than in the published FRM
- Extensions: FRM based on expectiles [Ren, Lu, Li & Härdle \(2022\)](#), for cryptocurrencies [Ren, Althof & Härdle \(2022\)](#) and for the EU energy sector [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)

GACV in a Calm and in a Crisis Window



- JPMorgan regressed on the other 12 banks and three lagged macro variables, 63 days; dotted line: selected λ ; orange: active coefficients
- Calm window (to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 of 15 coefficients active
 - ▶ COVID-19 window (to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 active
- In the crisis the whole curve is about three times higher: the check loss is in return units, and returns are larger
- With few regressors, the minimum lies where almost no penalty is applied: the selection is noisy

The Financial Risk Meter of Thirteen Banks

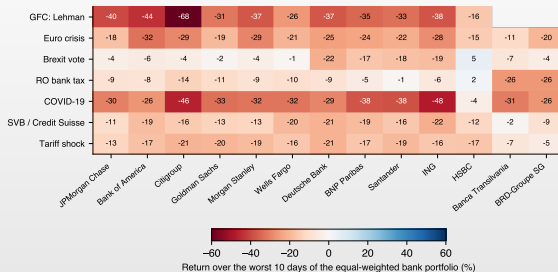


- Top: average GACV penalty over 13 banks; bottom: share of active predictors (12 banks and 3 lagged macro variables) when λ is held fixed, with the VIX; 390 windows of 63 days
- The average GACV penalty is noisy; its correlation with the VIX is only 0.35
 - With 15 regressors and 63 days the fit hardly needs a penalty; the published FRM uses about 100 firms, so that the penalty must do real work
- At a fixed penalty $\lambda = 2.3 \times 10^{-4}$, the share of active predictors (bank links and macro variables) follows the VIX more closely: correlation 0.62
 - From 31% (2017-11-21) to 68% (2020-06-17); average 42%
- Lesson: this small-panel application does not show that either the selected penalties or a fixed-penalty count of active predictors is a reliable systemic-risk indicator; the fixed-penalty count is only an exploratory sensitivity check

Stress Tests and Historical Scenarios

- A stress test asks how much we would lose in a **severe but plausible** scenario: **historical** (replay a past crisis), **hypothetical** (shock chosen factors), **reverse** (find the most plausible scenario behind a fatal loss)
 - ▶ Supervisory tests of large banks: [Federal Reserve stress tests](#) in the US, EU-wide tests of the [European Banking Authority](#); global systemically important banks hold extra capital [Basel Committee \(2018\)](#)
- Seven episodes, each a window of a few weeks around a known shock
 - ▶ Lehman (autumn 2008), the euro crisis (summer 2011), the Brexit vote (June 2016), the Romanian bank tax (December 2018), COVID-19 (March 2020), SVB and Credit Suisse (March 2023), the tariff shock (April 2025)
- Inside each window we find the **10 consecutive trading days with the lowest cumulative return** of the equal-weighted bank portfolio and report every bank's return over that interval
- The 2008 episode uses the 11 international banks only: the Romanian series start in 2010
- Ten days match the horizon of market-risk capital ([Chapter 7](#))

Seven Historical Stress Scenarios

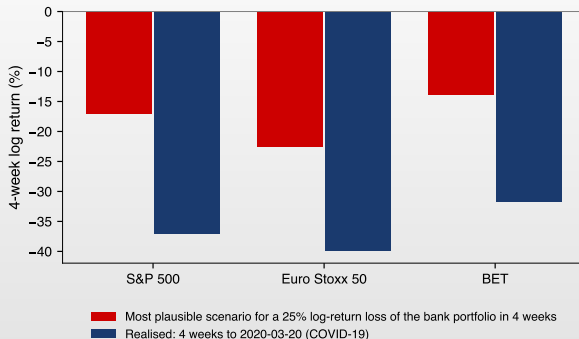


- Return (%) of each bank over the worst 10 consecutive days of the equal-weighted bank portfolio in each episode; blank: no data
- Portfolio loss over the worst 10 days: Lehman 37.6% (2008-11-10 to 2008-11-21), COVID-19 31.9%, euro crisis 22.3%
 - Tariffs 16.0%, SVB 14.8%, Romanian bank tax 10.2%, Brexit 7.9%
- Worst single cells: Citigroup -68% in 2008, ING -48% in March 2020
- Shocks are local as well as global: Brexit hit mainly European banks; the Romanian bank tax hit Romanian banks hardest (-26% and -26%), during a December 2018 sell-off that also cost US banks -8% to -14%
- A stress test built only on global crises would miss the domestic political shock

Reverse Stress Testing

- Factor model: weekly bank log returns (%) $X_i = \beta_i^\top f + e_i$, factors $f = (\text{S\&P 500, Euro Stoxx 50, BET})$, over h weeks $f \sim N(0, h\Sigma)$ (no serial covariance of weekly factor returns)
 - ▶ β_i : the bank's factor betas; e_i : its idiosyncratic return; Σ : weekly covariance matrix of the factors
 - ▶ Loss of the equal-weighted portfolio in log-return points: $L = -b^\top f$, with b the average of the β_i ; a factor-only approximation that sets the residual $\sum_i w_i e_i$ to zero
- The most plausible scenario with loss ℓ^* minimises the Mahalanobis distance $f^\top (h\Sigma)^{-1} f$ subject to $-b^\top f = \ell^*$
 - ▶ $f^* = -\ell^* \frac{h\Sigma b}{b^\top h\Sigma b}, \quad d^* = \frac{\ell^*}{\sqrt{b^\top h\Sigma b}}$
- d^* counts portfolio standard deviations: the larger it is, the less plausible the loss under the model
- A two-factor example worked step by step: [▶ Appendix](#) and Seminar 18, A7

The Most Plausible Scenario for a 25% Log-Return Loss



- Weekly data 2010–2026 (871 weeks), four-week horizon; red: f^* for a log-return loss $\ell^* = 25\%$; blue: factor moves in the worst four weeks of 2020
- Portfolio betas: $b = (0.45, 0.65, 0.18)$; four-week standard deviation 5.79%
 - ▶ A log-return loss of 25% (an arithmetic loss of 22.1%) needs log returns of S&P 500 -17.2% , Euro Stoxx 50 -22.6% , BET -13.9% together
- Distance $d^* = 4.31$: under the Normal model its probability is 8.0×10^{-6} per four weeks, practically impossible
- Yet in the four weeks to 2020-03-20 the portfolio lost 52.0% in log terms, with factor falls larger than f^* (the model implies 48.5%)
- The shape of the scenario is informative; its probability is not: heavy tails and rising correlations make Normal probabilities far too small

Climate Risk and the ECB Climate Stress Test of 2022



Seat of the European Central Bank, Frankfurt,
2015

Photo: DXR (2015); CC BY-SA 4.0; Wikimedia
Commons

- **Physical risk:** losses from floods, droughts, heat and storms that damage collateral and borrowers
 - ▶ **Transition risk:** losses from the move to a low-carbon economy (carbon prices, regulation, technology)
- “Green swan” risks: potentially extremely disruptive events that could be behind the next systemic financial crisis; backward-looking risk models cannot anticipate them [Bolton et al. \(2020\)](#)
- 104 significant banks took part; a bottom-up projection of losses covered 41 of them [ECB Banking Supervision \(2022\)](#)
- About 60% of banks had no climate risk stress-testing framework; only 20% considered climate risk when granting loans
- Credit and market losses of around EUR 70 billion for the 41 banks, under a short-term disorderly transition and two physical risk scenarios
 - ▶ The ECB states that this significantly understates the risk: data gaps, rudimentary models, no second-round effects

Limits of Stress Tests

- ▣ Scenarios are chosen by people: the crisis that comes is rarely the one that was tested
- ▣ Second-round effects (fire sales, funding runs, counterparty losses) are hard to model and usually left out, as the ECB noted for its climate test
- ▣ Market-based measures (MES, CoVaR, connectedness) react quickly but only after prices move; balance-sheet stress tests look forward but slowly
- ▣ Early-warning research combines both, e.g. entropy-based measures of market risk [Pele, Lazar & Dufour \(2017\)](#)

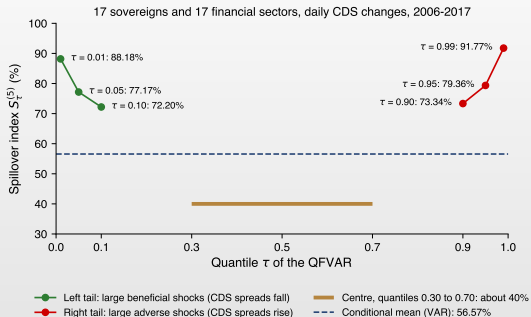
Case Study: Question and Motivation

- Ando, Greenwood-Nimmo & Shin (2022) ask whether the topology of a financial network changes with the **size** of the shocks that hit it (Section 1)
 - ▶ Their version of the question of Diebold & Yilmaz (2014): how much of the future uncertainty of i is due to idiosyncratic shocks from j , as the shock size varies?
- Why it matters: the connectedness of this chapter is estimated at the conditional mean (OLS, LASSO, elastic net Demirer et al. (2018)), i.e. it describes how an average shock spreads
 - ▶ Systemic shocks are larger than average; reading a rise in rolling connectedness as a crisis signal assumes that mean relations also hold in the tails (Section 1)
- Link to this chapter: the quantile regression of CoVaR Adrian & Brunnermeier (2016), Koenker & Bassett (1978), now inside the VAR and its variance decomposition
 - ▶ Giglio, Kelly & Pruitt (2016) showed that systemic risk shifts the lower tail of future output more than its centre; Ando, Greenwood-Nimmo & Shin (2022) ask the same tail-versus-centre question of the network itself
- Application: the credit risk loop between sovereigns and their banks, in which bailouts move bank risk onto the state (Sections 1 and 3)

Case Study: Data and Method

- Data (Section 3): daily first differences of 5-year CDS spreads (basis points) of 17 sovereigns and 17 national financial sectors, $m = 34$, 3 January 2006 – 29 December 2017, 3 129 days
- Quantile factor VAR (QFVAR), Eqs. (5)–(7): $y_t = \mu + \sum_{j=1}^p \Phi_j y_{t-j} + \Lambda f_t + v_t$, each equation fitted at quantile τ by minimising the check loss
 - ▶ y_t : the m CDS changes; μ : constants; Φ_j : VAR coefficient matrices; Λ : factor loadings
 - ▶ Unobserved factors f_t absorb common shocks, so v_t is idiosyncratic; their number by the criterion of [Ando & Bai \(2020\)](#), Eq. (8): 10 at the mean (Figure 1)
- Generalized forecast error variance decomposition (FEVD) of the idiosyncratic shocks at quantile τ , Eq. (17), rows normalised to 1; spillover index $S_\tau^{(h)} = m^{-1} \sum_i F_{i \leftarrow \bullet, \tau}^{(h)}$, Eq. (20)
 - ▶ $F_{i \leftarrow \bullet, \tau}^{(h)}$: the share of the h -day forecast error variance of i that comes from the others, the “from” of the connectedness table; $S_\tau^{(h)}$ is its average over the m series
- Rolling windows of $w = 250$ days (2 880 windows), number of factors fixed at its full-sample value; relative tail dependence $\text{RTD} = S_{0.95}^{(5)} - S_{0.05}^{(5)}$ (Section 4.3)
 - ▶ For CDS changes the right tail ($\tau = 0.95$) holds large adverse shocks, the left tail ($\tau = 0.05$) large beneficial ones
- Full specification (factor criterion, lag order, horizon) and the complete results: ▶ Appendix

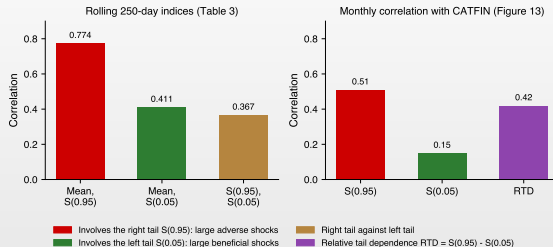
Case Study: Spillovers across Quantiles



- Full-sample spillover index $S_\tau^{(5)}$ of the QFVAR at each quantile τ against the conditional-mean index; source: [Ando, Greenwood-Nimmo & Shin \(2022\)](#), Figure 4
- The profile is U-shaped: about 40% in the centre, 88.18% at $\tau = 0.01$ and 91.77% at $\tau = 0.99$, more than twice the central level
 - The mean index (56.57%) lies between the two regimes and describes neither: too high for small shocks, too low for large ones
- On the full sample the two tails are almost symmetric:
 - RTD = 79.36 – 77.17 = 2.19 percentage points
 - The asymmetry is stronger and varies over time in rolling windows (next chart)
- Consequence for stress testing: a scenario built on mean connectedness understates how much of the variance of a sector's 5-day forecast error comes from other sectors' idiosyncratic shocks in a crisis

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Case Study: The Tails Move Apart



- Left: correlations between the rolling indices ($w = 250$ days, $h = 5$); right: monthly correlations with CATFIN; source: Ando, Greenwood-Nimmo & Shin (2022), Table 3 and Figure 13
- The mean index follows the adverse tail (0.774 with $S_{0.95}$) far more closely than the beneficial tail (0.411 with $S_{0.05}$)
 - Movements of the mean index mirror mainly the adverse tail, so a rise in mean connectedness is read as a rise in bad-news spillovers
- The two tails share little variation (0.367): they are two indicators, not one index measured twice
 - Hence RTD is informative: in rolling windows it reaches about ± 15 percentage points, against 2.19 on the full sample
- Against CATFIN: 0.51 for $S_{0.95}$, 0.42 for RTD, only 0.15 for $S_{0.05}$
 - CATFIN is more strongly correlated with the intensity of adverse-shock spillovers than with that of beneficial-shock spillovers

Case Study: Replicate, Limits, Extend

- Replicate with the course data: daily closes of the 13 banks of this chapter (course dataset; US, euro area and United Kingdom, Romania), 2010–2026; y_t = daily loss $-r_t$ in %, so that the right tail is the adverse one, as for CDS
 - ▶ Data limitation to state: equity losses instead of CDS spreads, 13 banks and no sovereigns ($m = 13$ instead of 34), different closing times of US, European and Romanian markets
- Limits stated by the authors: the lag order of the mean is imposed at every quantile, and the quantile is held fixed along the horizon (Sections 2.1–2.2, footnote 6)
- Open extension
 - ▶ Does the RTD of the 13 banks rise before March 2020 and the March 2023 bank failures?
 - ▶ Does it predict bank stress beyond current volatility and the mean-based index of this chapter?
 - ▶ For forecasting, select lag order and factors on a training sample only

An Open Question

Does network connectedness measure systemic risk, or only volatility?

- Connectedness indices are monitored as early-warning tools, but an index built from a VAR on volatilities may co-move with volatility without adding predictive information
 - ▶ Live research: global bank networks from regularised VARs [Demirer et al. \(2018\)](#); connectedness in the tails differs from connectedness in the mean [Ando, Greenwood-Nimmo & Shin \(2022\)](#)
- Formal version, for future bank stress y_{t+h} , total connectedness C_t and current volatility σ_t :
$$y_{t+h} = a + \beta_C C_t + \gamma \ln \sigma_t + e_{t+h}, \quad H_0: \beta_C = 0, \quad H_1: \beta_C > 0$$
 - ▶ Evidence against: after controlling for current volatility, the confidence interval for β_C rules out economically relevant effects; non-significance alone is not enough
- Identification: a network edge is a share of forecast error variance, not a causal exposure
 - ▶ A common shock raises volatility and connectedness together, with no bank infecting another

The Discovery Loop, Step by Step

- **Literature:** map what is known, then verify every reference
 - ▶ Resolve each DOI on [Crossref](#) and check that the paper really controls for volatility; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing readings of a rise in connectedness, each with its own test
 - ▶ E.g. a common shock fades with volatility; contagion has a direction: shocks of net transmitters precede those of receivers
- **Data and code:** on the course data, replicating a known number first
 - ▶ Replicate first: mean 62.0% and maximum 79.3% (2020-08-27), as in this chapter
- **Robustness,** fixed before the predictive regressions are run
 - ▶ Every cell of the grid is reported; the family is corrected with Holm or controlled with the FDR
- **Critique:** ask the assistant to attack the result
 - ▶ Expect: add current volatility as a control, then test the increment of R^2 and of the out-of-sample loss
- **Report:** standardised effects with HAC t -statistics, every cell of the grid, what failed
 - ▶ The prompts that shaped the design go into `AI_USE.md`
- Reusable prompts for every step and the researcher's checklist: ▶ Appendix

Mini-Case: Is Connectedness More Than Volatility?

- A typical assistant proposal: “high total connectedness warns of future bank stress”; we test it on 731 weekly dates, 2011–2026
 - ▶ y_{t+h} : log mean Parkinson volatility of the 13 banks over the next $h = 63$ days; control: the same over the last 20 days; standardised variables, Newey–West with 13 lags
- Connectedness alone: $\hat{\beta}_C = 0.40$ ($t = 4.63$), $R^2 = 0.16$: the proposal looks right
- With current volatility: $\hat{\beta}_C = 0.08$ ($t = 0.75$), $\hat{\gamma} = 0.53$ ($t = 4.13$); $R^2 = 0.343$ vs 0.339 with volatility alone
 - ▶ Correlation of C_t with current volatility: 0.59; for the next 63-day maximum drawdown of the bank portfolio, $\hat{\beta}_C = -0.10$ ($t = -1.32$) even when connectedness is the only regressor
- Interpretation: once current volatility is controlled for, the additional predictive contribution of connectedness is not significant; this does not prove it is zero, but in our sample the persistence of volatility accounts for most of the predictive content; network information may still matter in the tails [Ando, Greenwood-Nimmo & Shin \(2022\)](#) or for the direction of transmission

Your Turn: a Project Seed

- **Question:** does **tail** connectedness predict bank distress beyond volatility, out of sample?
- **Replicate first:** our rolling index (mean 62.0%, spring 2020 78.4%), then one headline connectedness figure of [Demirer et al. \(2018\)](#) on their sample
 - ▶ Extend: quantile connectedness at $\tau = 5\%$ [Ando, Greenwood-Nimmo & Shin \(2022\)](#), out-of-sample evaluation from 2018, MES and SRISK as alternative targets
- **Deliverables:** a grid of specifications fixed in writing before the regressions are run, a notebook that regenerates every number, HAC inference and out-of-sample losses for every cell
- **AI log:** prompts that shaped the design in AI_USE.md; each wrong claim (a VAR on price levels, OLS t -statistics on overlapping windows) and the test that caught it in AI_ERRORS.md ([templates](#))

Which Bank Is Most Systemic?

Bank	VaR 1%	MES 5%	ΔCoVaR 1%	Net spillover	L^*
JPMorgan	4.83	3.22	4.62	9.8	7.0
Bank of America	5.99	3.85	4.60	10.9	6.2
Citigroup	5.87	4.03	3.81	19.0	6.0
Goldman Sachs	4.81	3.34	4.12	2.4	6.8
Morgan Stanley	5.83	4.04	4.00	9.9	6.0
Wells Fargo	5.78	3.32	4.89	3.6	6.8
Deutsche Bank	7.19	4.54	4.24	-12.6	6.1
BNP Paribas	6.18	4.53	4.84	6.0	6.1
Santander	5.83	4.23	4.42	-5.4	6.4
ING	7.16	4.81	4.42	5.2	5.8
HSBC	4.66	2.54	4.07	-23.6	8.3
Banca Transilvania	5.06	3.14	—	-12.2	6.5
BRD-Groupe SG	4.42	2.69	—	-13.0	7.1

- ▣ Daily losses in %; net spillover of volatility (to minus from); L^* : break-even leverage with $k = 8\%$
- ▣ Rank correlations across the eleven international banks
 - ▶ MES and ΔCoVaR : -0.01; MES and net spillover: 0.04; ΔCoVaR and net spillover: -0.01
 - ▶ VaR and MES: 0.88
- ▣ No single ranking: supervisors combine several indicators with size and balance-sheet data, as in the [Basel Committee \(2018\)](#) methodology
- ▣ Romanian banks: moderate MES, net receivers of volatility, break-even leverage close to that of US banks

Key Takeaways

- ▣ Systemic risk lives in the joint tail: exposure (MES, SRISK), contribution (ΔCoVaR) and connectedness (spillover and tail networks)
- ▣ European banks have the largest MES in our sample; US banks are the main net transmitters of volatility
- ▣ VaR and ΔCoVaR rank banks differently; rankings are uncertain: test differences, report intervals, state αn
- ▣ Market-based measures jump in crises; stress tests and balance sheets are needed to see risk building up
- ▣ Most of the volatility forecast error variance of Romanian banks comes from their own shocks; they receive more than they transmit

Common Pitfalls

- ▣ Reading ΔCoVaR as causal: a common shock creates the same pattern
- ▣ Reading overlapping intervals as “equal” and separate ones as “different”: test the difference on joint draws, and correct for the number of pairs
- ▣ Including the bank itself in the system portfolio: its own return adds a mechanical self-contribution to the link
- ▣ Orthogonal variance decompositions whose results depend on the order of the banks
- ▣ Reporting SRISK without balance-sheet data, or from the $1 - e^{-18\text{MES}}$ shortcut as if it were the published measure
- ▣ Drawing every Granger edge significant at 5% out of 156 tests, or every net label without its interval

Appendix: MES as an Euler Contribution: Proof

- Portfolio $X_p = \sum_i w_i X_i$ with a continuous distribution; $ES_\alpha(X_p) = -\mathbb{E}[X_p \mid X_p \leq q_\alpha(X_p)]$
 - ▶ ES_α is positively homogeneous of degree 1 in w , so by Euler's theorem $ES_\alpha(X_p) = \sum_i w_i \partial ES_\alpha / \partial w_i$
 - ▶ Differentiating $-\frac{1}{\alpha} \mathbb{E}[X_p 1\{X_p \leq q_\alpha(w)\}]$: the term in $\partial q_\alpha / \partial w_i$ is $-\frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i + \frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i = 0$
 - ▶ Hence $\partial ES_\alpha / \partial w_i = -\mathbb{E}[X_i \mid X_p \leq q_\alpha(X_p)] = MES_{i,\alpha}$ with $m = p$ [Gouriéroux, Laurent & Scaillet \(2000\)](#), [Scaillet \(2004\)](#)
- Check on our data: equal-weighted portfolio of the six US banks, $\alpha = 5\%$ (198 tail days)
 - ▶ $ES_{5\%}$ of the portfolio = 4.32% = average of the six MES conditional on the portfolio (4.32%): exact, also for the historical estimator
 - ▶ Conditioning on the S&P 500 instead gives an average MES of 3.63% (JPMorgan: 3.22% vs 3.85%): the identity fails when $m \neq p$
- Under the bivariate Normal distribution with zero means, $MES_{i,\alpha} = \beta_i ES_\alpha(X_m)$: a beta times a constant (Seminar 18, A3)

Appendix: LRMES Simulation: Innovations and Recursions

- Standardised residuals $\epsilon_{it} = r_{it}/\sigma_{it}$, $\epsilon_{mt} = r_{mt}/\sigma_{mt}$; S is their sample correlation matrix, the long-run target of Q_t
 - ▶ Bank innovation orthogonal to the market: $\xi_{it} = (\epsilon_{it} - \rho_t \epsilon_{mt})/\sqrt{1 - \rho_t^2}$ [Brownlees & Engle \(2017\)](#), Appendix A
- Each simulated day s : draw a past date τ , take the pair $(\xi_{i\tau}, \epsilon_{m\tau})$ and rebuild the bank shock with the **simulated** correlation
 - ▶ $\epsilon_{i,t+s} = \rho_{t+s} \epsilon_{m\tau} + \sqrt{1 - \rho_{t+s}^2} \xi_{i\tau}$, then update σ^2 and Q with the new returns
- Restrictions: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$, $\alpha + \gamma/2 + \beta < 1$ (finite variance for symmetric innovations; in general $\alpha + \gamma \mathbb{E}[\epsilon^2 \mathbf{1}\{\epsilon < 0\}] + \beta < 1$); $a, b > 0$, $a + b < 1$, so that $\sigma_t^2 > 0$ and Q_t is positive definite
 - ▶ Initialisation: $Q_1 = S$ in estimation; the simulation starts from σ_{T+1}^2 and Q_{T+1} filtered to the last date

Appendix: Quantile Regression: Asymptotics

- Asymptotics for a fixed α , with $\hat{\theta}_\alpha = (\hat{a}_\alpha, \hat{b}_\alpha)^\top$, n days, $x_t = (1, X_{i,t})^\top$ and f_t the conditional density of Y at its α -quantile
 - ▶ $\sqrt{n}(\hat{\theta}_\alpha - \theta_\alpha) \Rightarrow N\left(0, \alpha(1 - \alpha) D_1^{-1} D_0 D_1^{-1}\right)$, $D_0 = \mathbb{E}[x_t x_t^\top]$, $D_1 = \mathbb{E}[f_t x_t x_t^\top]$
 - ▶ Assumes independent observations (or scores $x_t(\alpha - 1\{u_t < 0\})$ that are a martingale difference), $f_t > 0$ near the quantile and D_0 , D_1 positive definite; with serially dependent scores D_0 needs a long-run (HAC) estimate or a block bootstrap
 - ▶ The **sparsity** $1/f_t$ must be estimated (kernel or bandwidth); at $\alpha = 1\%$ few observations lie near the quantile, so this estimate is noisy
- The slope b_α can change with α : the bank may matter more for the system in the tail than in the centre; least squares gives only the mean slope

Appendix: Extremal Quantiles: Why $\alpha = 1\%$ Is Hard

- The effective tail sample is αn : 40 days at 1% and 198 at 5% out of 3 960
 - ▶ The Normal approximation above assumes $\alpha n \rightarrow \infty$ fast; with αn of a few dozen it can be poor, and extremal-quantile theory (extreme-value limits, subsampling) is the alternative built for this case [Chernozhukov & Fernández-Val \(2011\)](#)
- In our data the block-bootstrap standard error of $\Delta\text{CoVaR } 1\%$ is about 0.6 times the kernel-based standard error (median over banks): the two methods give clearly different uncertainty
 - ▶ The kernel-based standard error rests on a sparsity estimate from the few observations near the 1% quantile; the bootstrap redraws the same few tail days: neither is reliable when αn is this small
- Practical rule: report $\alpha = 5\%$ and $\alpha = 1\%$ with αn ; expect the 1% ranking to be the noisier one
 - ▶ The low Spearman correlation between the 1% and 5% rankings (0.10) may reflect estimation noise, genuine differences between the two tail levels, or both

Appendix: Testing ΔCoVaR : Three Hypotheses

- (1) No contribution, $H_0: b_\alpha = 0$: Wald test with the sandwich variance of quantile regression [Koenker & Bassett \(1978\)](#)
 - ▶ Tests of linear hypotheses across quantiles use the joint asymptotics of $(\hat{b}_\alpha, \hat{b}_{50\%})$ [Koenker & Bassett \(1982\)](#)
- (2) The tail link equals the central link, $H_0: b_\alpha = b_{50\%}$: is the bank special in crises?
 - ▶ We estimate the variance of $\hat{b}_\alpha - \hat{b}_{50\%}$ from 999 moving-block bootstrap draws that keep the pairs of days together
- (3) Equal contributions, $H_0: \Delta\text{CoVaR}_i = \Delta\text{CoVaR}_j$: the same resampled days for **all** banks give the joint distribution
 - ▶ $z_{ij} = (\widehat{\Delta\text{CoVaR}}_i - \widehat{\Delta\text{CoVaR}}_j) / \widehat{\text{se}}^*(\cdot)$ for each pair; Holm correction over the pairs [Holm \(1979\)](#); a joint Wald test that all are equal
 - ▶ Significance and dominance tests for systemic importance: [Castro & Ferrari \(2014\)](#)

Appendix: Testing ΔCoVaR 1%: Results

Bank	ΔCoVaR	$\hat{b}_{1\%}$	z	$\hat{b}_{50\%}$	$\hat{b}_{1\%} - \hat{b}_{50\%}$	p
JPMorgan	4.62	0.94	6.9	0.96	-0.02	0.66
Bank of America	4.60	0.76	6.5	0.78	-0.02	0.73
Citigroup	3.81	0.64	4.7	0.76	-0.11	0.00
Goldman Sachs	4.12	0.85	5.6	0.84	0.00	0.99
Morgan Stanley	4.00	0.68	4.5	0.73	-0.05	0.28
Wells Fargo	4.89	0.84	4.3	0.78	0.06	0.17
Deutsche Bank	4.24	0.59	4.1	0.60	-0.02	0.67
BNP Paribas	4.84	0.77	6.5	0.74	0.03	0.37
Santander	4.42	0.75	4.5	0.69	0.07	0.07
ING	4.42	0.61	4.4	0.71	-0.10	0.00
HSBC	4.07	0.87	4.0	0.86	0.01	0.96

- z : kernel (sandwich) t -ratio of $\hat{b}_{1\%}$, valid only for serially independent scores: every bank contributes; p : bootstrap test of $b_{1\%} = b_{50\%}$, significant for 2 of 11 banks (2 after Holm)
- ▶ Equal contributions within a region: 4 of 25 pairs differ at 5%, 0 after Holm; joint Wald $p = 0.01$ (US), 0.31 (Europe)
 - ▶ At 5%: 3 of 25 pairs differ, 0 after Holm; joint Wald $p = 0.014$ (US), 0.046 (Europe)

Appendix: Time-Varying ΔCoVaR with State Variables

- State variables M_{t-1} : log VIX, the return of the regional index and of the system, all lagged one day [Adrian & Brunnermeier \(2016\)](#)
 - ▶ $q_\alpha(X_{i,t} \mid M_{t-1}) = a_i + c_i^\top M_{t-1}$ at α and at 50% (a_i, c_i : intercept and coefficients of the bank's quantile regression)
 - ▶ $q_\alpha(X_t^{\text{sys}} \mid X_{i,t}, M_{t-1}) = a + b X_{i,t} + c^\top M_{t-1}$
- Then $\Delta\text{CoVaR}_{\alpha,t} = \hat{b} (\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i))$
 - ▶ It moves with the bank's conditional VaR, which moves with the VIX
- The slope $\hat{b}$ is constant: time variation comes from the state, not from the link

Appendix: Backtesting CoVaR and MES

- Systemic measures are conditional risk measures, so they need **conditional** backtests [Banulescu-Radu, Hurlin, Leymarie & Scaillet \(2021\)](#)
 - ▶ For $\text{CoVaR}^{\leq}$: hit $I_t = 1\{X_t^{\text{sys}} \leq -\text{CoVaR}_t^{\leq}, X_{i,t} \leq -\text{VaR}_{i,t}\}$ with $P(I_t = 1) = \alpha^2$ under H_0 if both the bank VaR and the conditional quantile are calibrated; Kupiec and Christoffersen tests on I_t ([Chapter 8](#)), plus a separate check of the bank VaR rate
 - ▶ For MES: the cumulative joint violation process of [Banulescu-Radu, Hurlin, Leymarie & Scaillet \(2021\)](#)
- Our out-of-sample check: historical $\text{CoVaR}_{5\%}^{\leq}$ on an expanding window from day 1 000, 2 960 forecast days per bank
 - ▶ Expected joint hits 7.4; observed 5–9 across the eleven banks (JPMorgan 9); smallest Kupiec $p = 0.35$: no rejection
 - ▶ Power is limited: if the true joint rate were twice α^2 , the 5% Kupiec test would reject in 62% of samples of this length with independent hits, so even a doubling is missed about one time in three

Appendix: VAR Assumptions and the Lag Order of the Rolling Index

- Vector autoregression (VAR) of order p for N series y_t : $y_t = c + \sum_{l=1}^p A_l y_{t-l} + \varepsilon_t$, $\text{Cov}(\varepsilon_t) = \Sigma$
 - ▶ Assumptions: stable VAR (roots of $\det(I - \sum_l A_l z^l)$ outside the unit circle), ε_t serially uncorrelated with finite, positive definite Σ
 - ▶ Moving-average form: $y_t = \mu + \sum_{h \geq 0} \Phi_h \varepsilon_{t-h}$, $\Phi_0 = I$, $\Phi_h = \sum_{l=1}^p A_l \Phi_{h-l}$ ($\Phi_h = 0$ for $h < 0$); e_i : the i -th unit vector
- Lag order: $p = 2$ is held fixed in every window so that the index does not jump when the selected order changes; BIC picks $p = 1$ in 100% of the windows, and with $p = 1$ the index has correlation 0.99 with the one shown

Appendix: Regularised VARs: the Elastic Net

- [Demirer et al. \(2018\)](#), Eq. (9): each VAR equation by the adaptive elastic net, λ chosen equation by equation by 10-fold cross-validation
 - ▶ $\hat{\beta} = \arg \min_{\beta} \sum_t (y_t - \sum_i \beta_i x_{it})^2 + \lambda \sum_i w_i (\frac{1}{2} |\beta_i| + \frac{1}{2} \beta_i^2)$, $w_i = 1/|\hat{\beta}_{i,OLS}|$
 - ▶ x_{it} : the lagged series (regressors); λ : penalty strength; w_i : adaptive weights, larger for coefficients small in OLS; the $|\beta_i|$ part sets coefficients to zero, the β_i^2 part shrinks them
 - ▶ Rolling windows of 150 days with the plain elastic net, $w_i = 1$ (Section 5, footnote 15); forecast horizon $H = 10$
- Why: selection and shrinkage in the VAR coefficients, without forcing a sparse **network** (links also come through Σ)
 - ▶ The paper does not state the lag order in the text; we keep the BIC order of our OLS VAR, $p = 2$
- Full sample, adaptive elastic net: $C = 63.3\%$ (OLS 63.9%); 59% of the coefficients are non-zero
 - ▶ With about 4 000 days the penalty hardly matters; it matters in short windows (rolling-window chart in the main text)

Appendix: Granger-Causality Networks and Multiple Testing

- Billio, Getmansky, Lo & Pelizzon (2012) draw an edge $j \rightarrow i$ when j Granger-causes i at 5%; with 13 banks this is $13 \times 12 = 156$ tests
 - ▶ Even with no links at all, about 7.8 edges would appear by chance
- Our tests: bivariate regressions, one lag, heteroskedasticity-robust Wald statistics
 - ▶ Daily returns: 57 edges at 5%, 33 after Holm [Holm \(1979\)](#), 55 with the false discovery rate controlled at 5% [Benjamini & Hochberg \(1995\)](#)
 - ▶ Log volatility, highly persistent: one extra lag that is not tested [Toda & Yamamoto \(1995\)](#), as in the lag-augmented tests of [Shi, Hurn & Phillips \(2020\)](#): 153, 150 and 153 edges
- Holm controls the chance of any false edge under arbitrary dependence; the FDR is the **expected** share of false edges among those drawn, and BH guarantees it for independent tests [Benjamini & Hochberg \(1995\)](#), while tests on overlapping bank pairs are dependent
- Daily returns of banks in different time zones: many “causal” edges only reflect that New York closes after Frankfurt

Appendix: Worked Example: Reverse Stress Test with Two Factors

- Betas $b = (1.2, 0.8)$; four-week covariance of the factors $h\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix} (\%^2)$; target loss $\ell^* = 20\%$
 - ▶ $h\Sigma b = (1.2 \cdot 16 + 0.8 \cdot 6, ; 1.2 \cdot 6 + 0.8 \cdot 9) = (24, 14.4)$
 - ▶ $b^\top h\Sigma b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$, standard deviation 6.35%
- Most plausible scenario and its distance
 - ▶ $f^* = -20 \cdot (24, 14.4)/40.32 = (-11.9\%, -7.1\%)$, $d^* = 20/6.35 = 3.15$
 - ▶ Normal probability of a loss at least this large: $\Phi(-3.15) = 0.08\%$ per four weeks
- The scenario loads more on the factor with the larger beta and variance, and both factors fall together because they are correlated

Appendix: Case Study: Data and Method

- Data (Section 3): daily first differences of 5-year CDS spreads (basis points) of 17 sovereigns and 17 national financial sectors, $m = 34$, 3 January 2006 – 29 December 2017, 3 129 days
 - ▶ A financial sector = equal-weighted CDS of its banks and insurers with assets of at least USD 10 billion, 199 institutions in all
- Quantile factor VAR (QFVAR), Eqs. (5)–(7): $y_t = \mu + \sum_{j=1}^p \Phi_j y_{t-j} + \Lambda f_t + v_t$, each equation fitted at quantile τ by minimising the check loss
 - ▶ Unobserved factors f_t absorb common shocks, so v_t is idiosyncratic; their number by the criterion of [Ando & Bai \(2020\)](#), Eq. (8): 10 at the mean (Figure 1)
 - ▶ Residual pairs with $|\rho| < 0.25$: 91.6% with factors, fewer than half without (Figure 2)
- Generalized FEVD of the idiosyncratic shocks at quantile τ , Eq. (17), rows normalised to 1; spillover index $S_\tau^{(h)} = m^{-1} \sum_i F_{i \leftarrow \bullet, \tau}^{(h)}$, Eq. (20)
 - ▶ VAR(1) chosen by BIC at the mean and kept at every quantile; $h = 5$ days; the quantile is held fixed along the forecast horizon (Sections 2.2 and 4)
- Rolling windows of $w = 250$ days (2 880 windows), number of factors fixed at its full-sample value; relative tail dependence $\text{RTD} = S_{0.95}^{(5)} - S_{0.05}^{(5)}$ (Section 4.3)
 - ▶ For CDS changes the right tail ($\tau = 0.95$) holds large adverse shocks, the left tail ($\tau = 0.05$) large beneficial ones

Appendix: Case Study: Key Results

- Full-sample spillover index (Figure 4): about 40% over the central 40% of the distribution; 72.20%, 77.17%, 88.18% at $\tau = 0.10, 0.05, 0.01$; 73.34%, 79.36%, 91.77% at $\tau = 0.90, 0.95, 0.99$; 56.57% at the mean
 - ▶ Small shocks stay local; large shocks, good or bad, spread; the mean averages the two regimes into one number
- Of the 1122 bilateral spillovers, 568 are below 1% at the mean and 829 at the median (Figure 7)
 - ▶ Away from the mean, the network has many weak links rather than a few strong ones; in the tails these weak links add up to strong average spillovers
- Rolling windows (Table 3): the mean index correlates 0.774 with $S_{0.95}$ but 0.411 with $S_{0.05}$; the two tails correlate only 0.367
 - ▶ The tails move apart: RTD is 2.19 percentage points on the full sample but reaches about ± 15 percentage points in rolling windows (Section 4.3, Figure 11)
- Monthly correlation with CATFIN [Allen, Bali & Tang \(2012\)](#), the average of three VaR 1% estimates for financial stocks (Figure 13): 0.51 for $S_{0.95}$, 0.15 for $S_{0.05}$, 0.42 for RTD
 - ▶ RTD rises with the Bear Stearns fund losses of 2007 and falls after the US bank rescue programme and the negative rates of the ECB: a fragility signal the mean hides

Appendix: Case Study: Replication Specification and Limits Stated by the Authors

- Replicate with the course data: daily closes of the 13 banks of this chapter (course dataset; US, euro area and United Kingdom, Romania), 2010–2026; y_t = daily loss $-r_t$ in %, so that the right tail is the adverse one, as for CDS
 - ▶ Specification: VAR order by BIC, factors by Eq. (8), $h = 5$, $\tau \in \{0.05, 0.5, 0.95\}$ and the mean, $w = 250$, RTD
 - ▶ Data limitation to state: equity losses instead of CDS spreads, 13 banks and no sovereigns ($m = 13$ instead of 34), different closing times of US, European and Romanian markets
- Limits stated by the authors: the lag order of the mean is imposed at every quantile, and the quantile is held fixed along the horizon (Sections 2.1–2.2, footnote 6)
 - ▶ 250 days are too few to select factors window by window, so their number is fixed (Section 4.3, Appendix B); shorter windows leave too few effective tail observations
 - ▶ Only direct spillovers of idiosyncratic shocks are measured, not propagation through the common factors (footnote 2); Greece is left out for a gap in its CDS data (Section 3, Appendix A)

Appendix: The Discovery Loop, Reusable Prompts

- **Literature:** map what is known, then verify every reference
 - ▶ “Which papers test whether connectedness indices predict bank distress out of sample, beyond volatility? Give DOI, sample, horizon and controls.”
- **Hypotheses:** competing readings of a rise in connectedness, each with its own test
 - ▶ “Give competing interpretations of a rise in total connectedness (common shock, contagion, artefact of rolling windows) and a test that discriminates between them.”
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Fit a VAR(2) to the log Parkinson volatilities of the 13 banks of the course data on 250-day windows and compute the generalized total connectedness with $H = 10$.”
- **Robustness,** fixed before the predictive regressions are run
 - ▶ “Pre-register: horizons 21/63/126 days, volatility and VIX controls, windows 150/250/500 days, HAC lags at least the overlap, out-of-sample evaluation from 2018.”
- **Critique:** ask the assistant to attack the result
 - ▶ “Act as a referee: is the predictive power of connectedness just the persistence of volatility? Which regression would show it?”

Appendix: What the Researcher Must Check

- **Non-stationary inputs:** a VAR on price levels, or on trending series, produces spurious connectedness
 - ▶ Test: unit-root tests on every input; all eigenvalues of the companion matrix of the fitted VAR strictly inside the unit circle in every window
- **Ordering:** Cholesky decompositions change with the order of the banks
 - ▶ Test: permute the columns; the generalized table must not change
- **Look-ahead:** a window labelled by its first date, or a lag order chosen on the full sample
 - ▶ Test: the index at t must not move when data after t are deleted
- **Overlap:** rolling indices and forward outcomes overlap, so OLS t -statistics are inflated
 - ▶ Test: Newey–West lags at least equal to the overlap, or a block bootstrap
- **Dimension:** 13 banks and $p = 2$ give 27 coefficients per equation from 250 days; more banks need shrinkage [Demirer et al. \(2018\)](#)
 - ▶ Test: compare with a LASSO VAR; unstable edges signal estimation noise, not structure

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Financial Risk Meter for Emerging Markets
- ☐  Dynamic Crypto Networks
- ☐  Network Centrality

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_18

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